



Habib University - City Campus

Course Title: **Electric Circuits II Lab (EE213L)**

Final Exam

Name: Khuzaima Ali Khan

Student ID: RA

Time duration: 150 mins

Grade Table (for teacher use only)

Question	Max. Marks	Score
1A.	2	
1B.	6	
1C.	2	
2A.	4	
2B.	4	
2C.	2	
3A.	2	
3B.	0.5	
3C.	2.5	
3D.	5	
Total	30	

Instructions:

- Mobile phones must be turned off and put away.
- Use of lab manual is not allowed.
- All the work should be done on the given question paper.
- No food or drink, other than water, is to be brought into the examination hall.
- You are not allowed to communicate by word of mouth or otherwise with other candidates
- You are not allowed to leave your seat unless granted permission by an invigilator.
- For assessment you are required to upload file (with proper comments and explanation with labelled figures).

Hardware: Designing Active Low Pass Filter

Q1: The active filter is presented as follows:

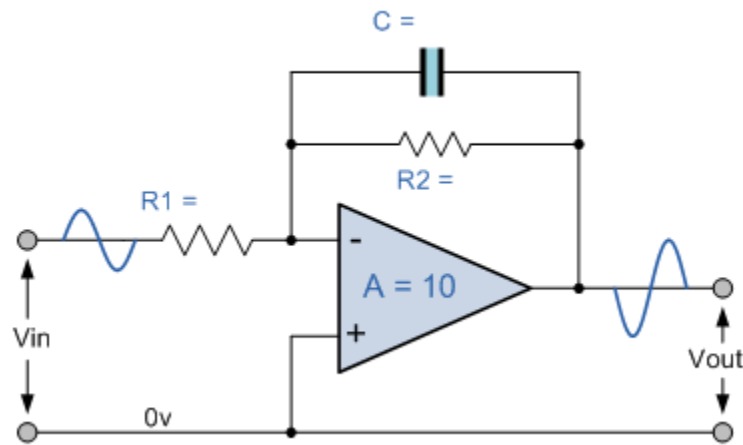


Figure 1: Schematic for Active low Pass filter

Hardware components required:

- LM741
- Resistors
- Capacitor

$$C_1 = 10 \text{ nF}$$

Given that:

$$f_c = 1 \text{ kHz}$$

$$A_F = 10$$

$$f_c = \frac{1}{2\pi R_2 C}$$

$$1k = \frac{1}{2\pi R_2 (10n)}$$

$$R_2 = 15.9 \text{ k}\Omega \approx 15 \text{ k}\Omega$$

$$DC \text{ Gain } (A_F) = -\frac{R_2}{R_1}$$

$$-10 = -\frac{15k}{R_1}$$

$$R_1 = 1.5 \text{ k}\Omega$$

- A. Calculate values for R_1 , R_2 and C to set gain = 10 V/V and cut-off frequency = 1KHz. **Note that,** For the sake of simplicity, we will not take into account the capacitance when calculating gain. (2)

$$R_1 = 1.5 \text{ k}\Omega$$

$$R_2 = 15 \text{ k}\Omega$$

$$C = 10 \text{ nF}$$

Following are the V_{in} parameters:

$$V_{amp} = 0.5 V_{p-p}$$

- B. Complete the following table: (6)

Table 1: Data Table for Low Pass Filter

Frequency (Hz)	10	100	500	1k	$f_c = 1010$	1.5K	5K	10k	50k	100K	500K	1M
Vin Max (V)	0.512	0.512	0.512	0.512	0.512	0.512	0.512	0.512	0.512	0.512	0.512	0.512
Vout Max (V)	5.12	5.12	4.48	3.36	3.36	2.6	0.94	0.496	0.104	0.052	0.022	0.052
$Af = \frac{V_{out \text{ Max}}}{V_{in \text{ Max}}}$	10	10	8.75	6.56	6.56	5.078	1.83	0.968	0.2031	0.1015	0.04296	0.1015
$gain(dB) = 20\log(A)$	20	20	18.840	16.34	16.34	14.1	5.27	-0.275	-13.844	-19.865	-27.3369	-19.86

```

f1 = [10    100    500    1e3    1010    1.5e3    5e3    10e3    50e3    100e3
500e3 1e6]

f1 = 1x12
    10    100    500    1000    1010    1500 ...

Vin1 = [0.512    0.512    0.512    0.512    0.512    0.512    0.512    0.512
0.512    0.512    0.512    0.512]

Vin1 = 1x12
    0.5120    0.5120    0.5120    0.5120    0.5120    0.5120    0.5120    0.5120 ...

Vout1 = [5.12    5.12    4.48    3.44    3.36    2.6    0.94    0.496    0.104
0.052    0.022    0.052]

Vout1 = 1x12
    5.1200    5.1200    4.4800    3.4400    3.3600    2.6000    0.9400    0.4960 ...

% finding gain
A_F = Vout1./Vin1

A_F = 1x12
    10.0000    10.0000    8.7500    6.7188    6.5625    5.0781    1.8359    0.9688 ...

gain_1 = 20*log10(A_F)

gain_1 = 1x12
    20.0000    20.0000    18.8402    16.5458    16.3414    14.1141    5.2772    -0.2758 ...

% plotting it
semilogx(f1,gain_1)
xlabel("frequency (Hz)")
ylabel("Gain (dB)")
grid on

```

- C. Plot the Bode plot of the filter showing cut-off frequency, pass band, and the 17 dB point. Using the semi-log graph provided with proper axes labelling and f_c . (2)

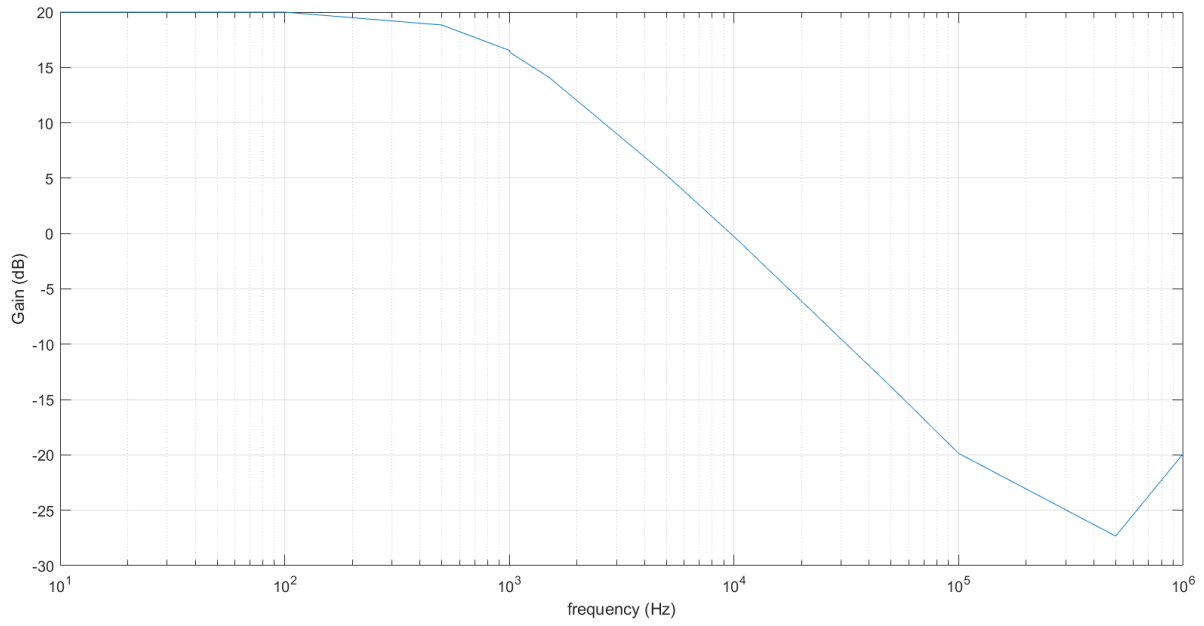


Figure 2: Gain(dB) Vs Frequency Graph for Low Pass Filter

MATLAB Simulink: To measure Voltage, Current and Power of three phase $\Delta - \Delta$ balanced system

Q2: The $\Delta - \Delta$ system is as follows:

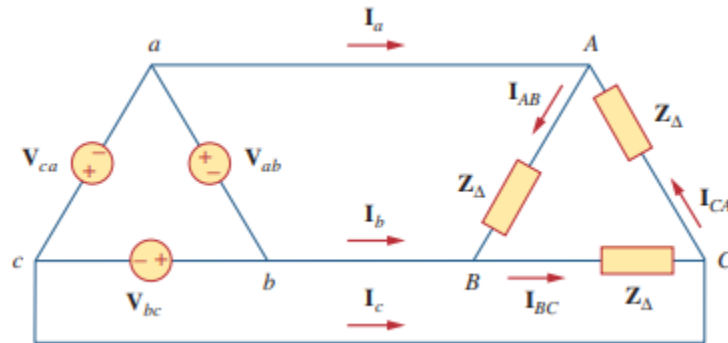


Figure 3: $\Delta - \Delta$ system

Connect a three-phase balanced load in delta connection with per phase load of $Z_{\Delta} = 100\Omega$, and $V_{ab} = 100\angle 0^\circ$ and $f = 50 \text{ Hz}$.

- A. Measure instantaneous Line and phase Voltage, Current and Power per phase. Also, measure total 3 phase power. Fill the following table: (4)

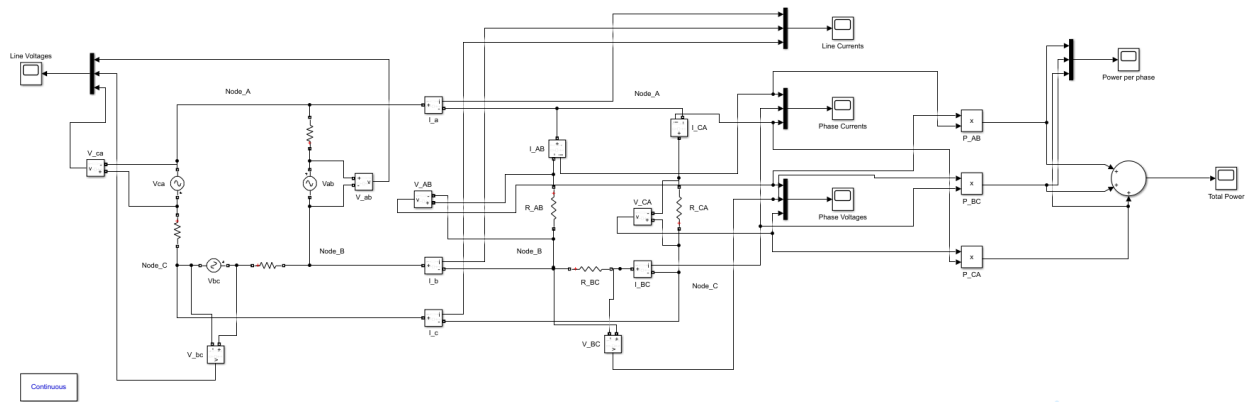
Table 2: Measured Values

Quantity	Peak Value
V_{ab}	100 V
V_{bc}	100 V
V_{ca}	100 V
I_a	1.72 A
I_b	1.72 A
I_c	1.72 A
I_{AB}	1 A
I_{BC}	1 A

I_{CA}	1 A
$P_{1\phi}$	98 W
P_{Total}	147 W (constant value)

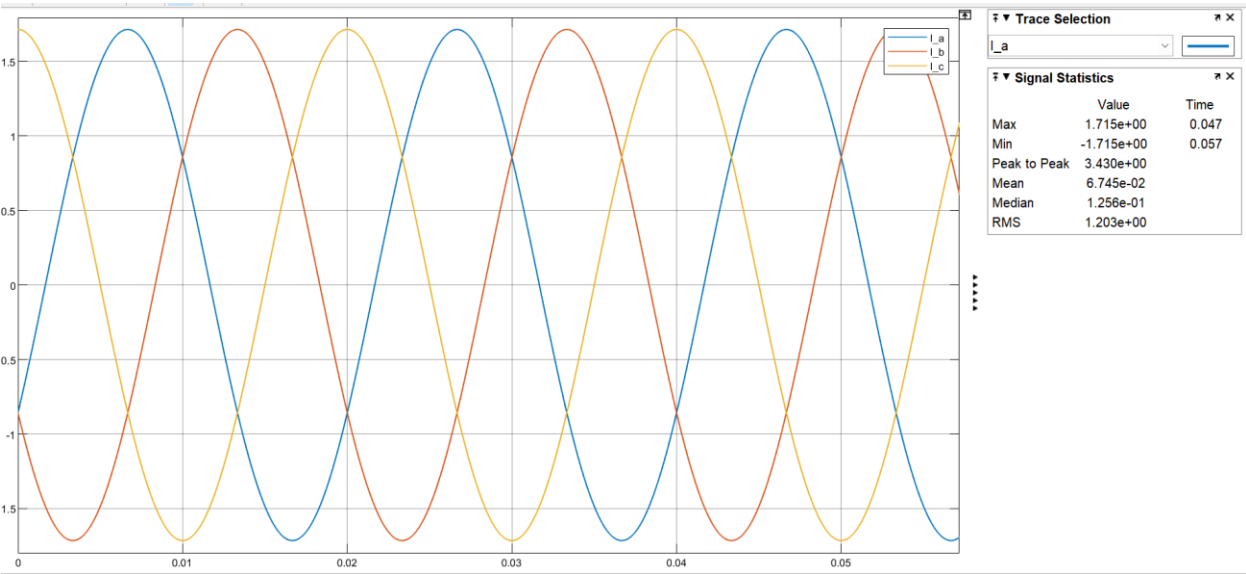
B. Attach screenshot of Simulink model & waveforms in report. Show all Peak & RMS values. (4)

Simulink Model

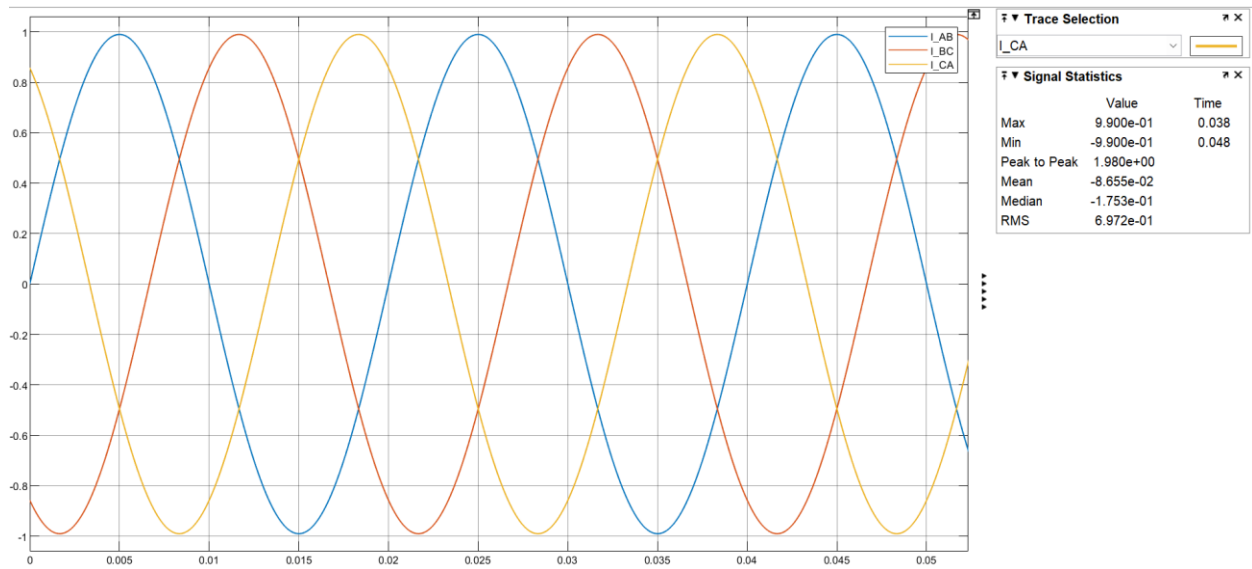


Waveforms

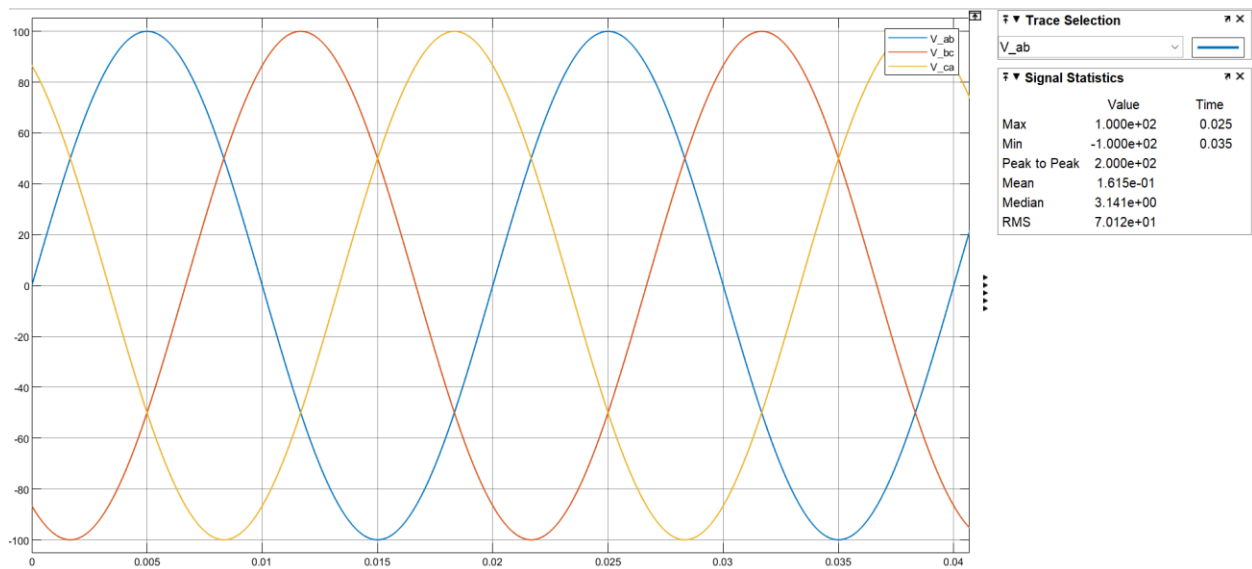
Line Currents:



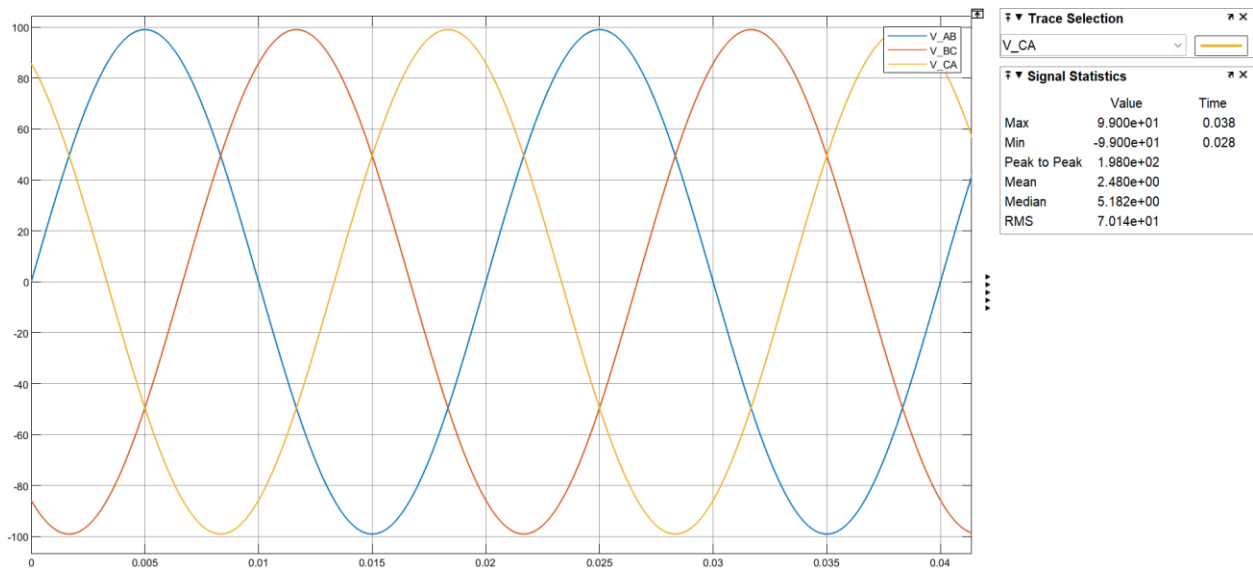
Phase Currents:



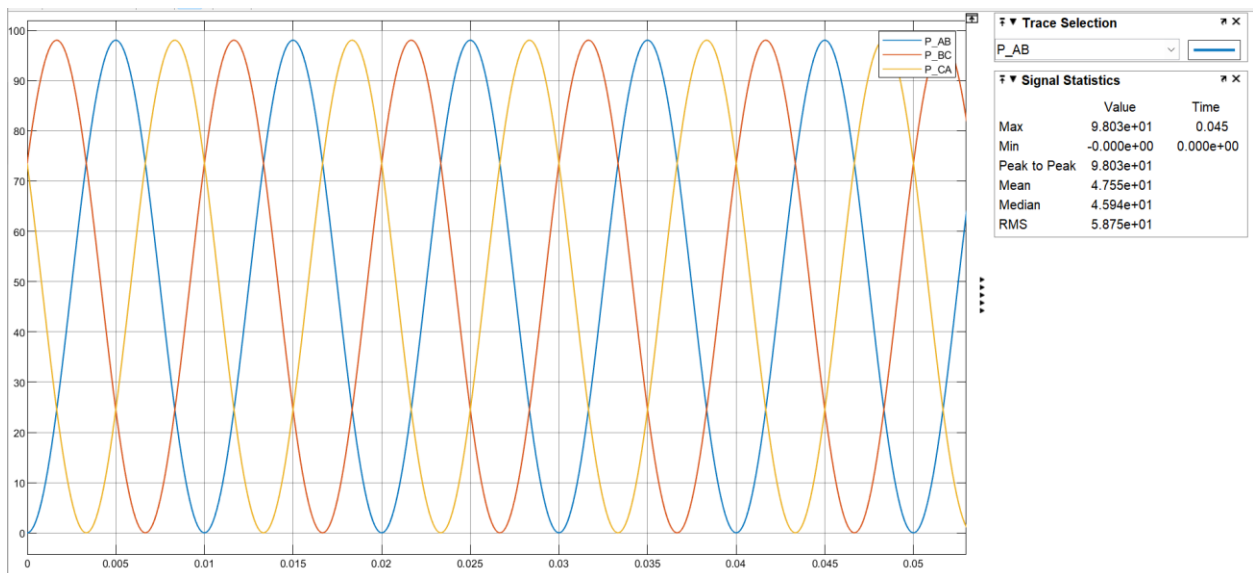
- Line Voltages:



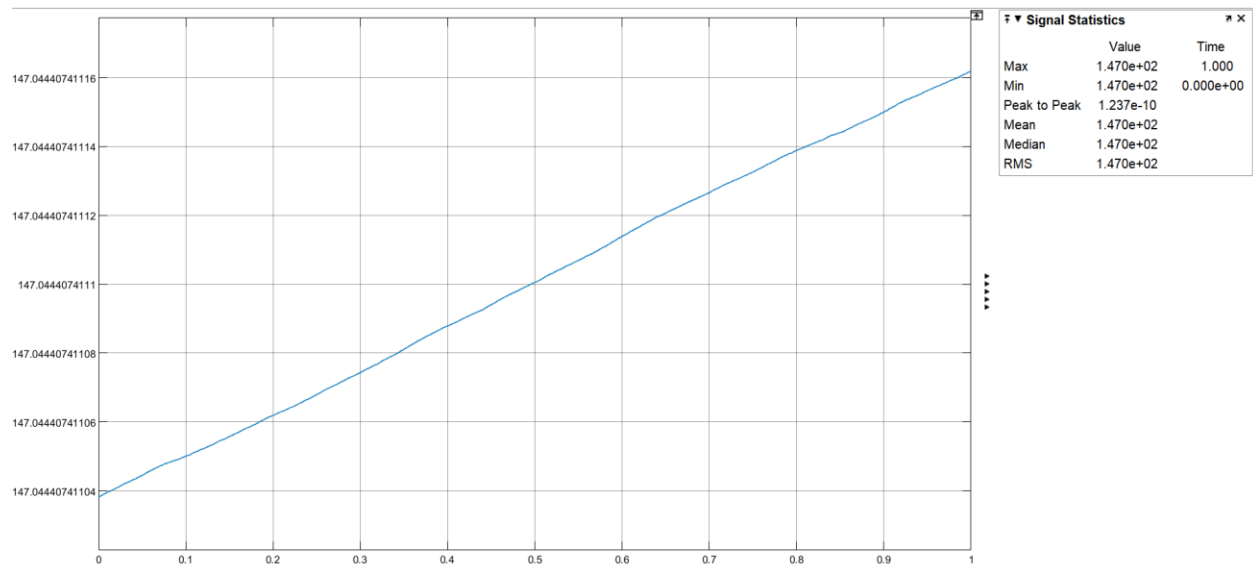
- Phase Voltages:



- **Power per phase:**



- **Total Power:**



C. Analyse the waveforms obtained. (2)

$$V_{line} = V_{\phi} = 100 \text{ V}$$

$$I_{line} = \sqrt{3} \cdot I_{\phi}$$

$$P_{1\phi}(Peak) = V_{\phi} I_{\phi} \approx 100 \text{ W}$$

P-Spice: Notch Filter Designing and Simulation

Refer to the following figure for the filter description:

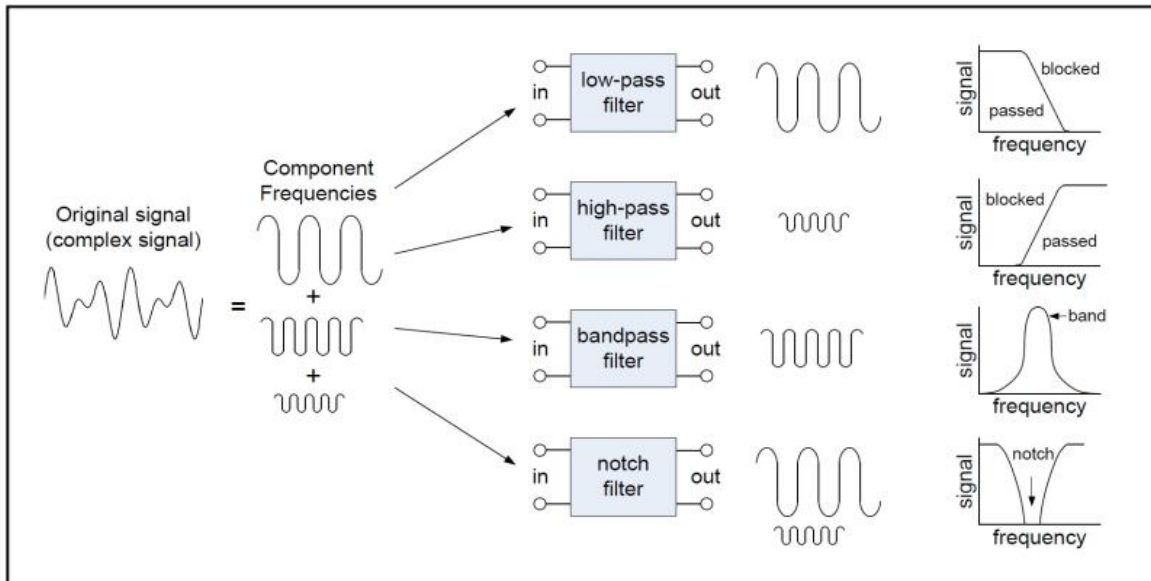


Figure 4: Types of filters

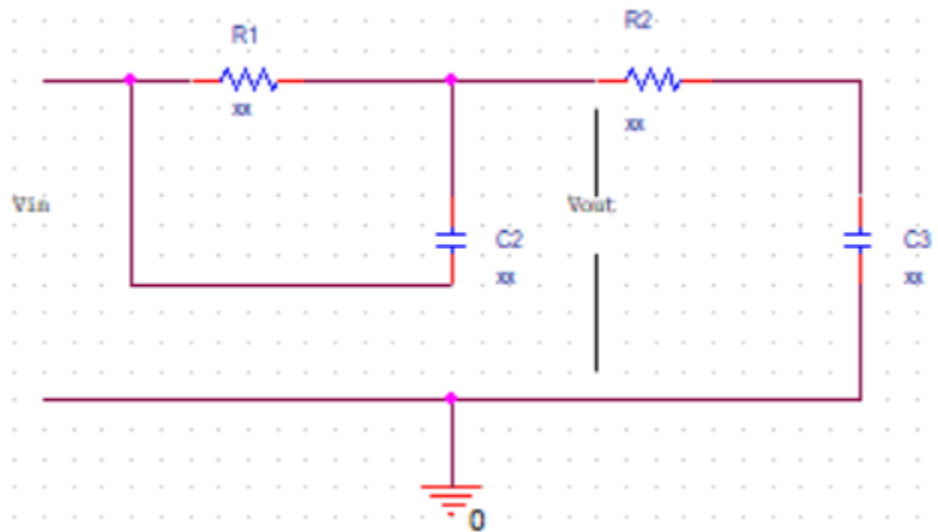


Figure 5: Circuit Diagram of Passive Notch Filter

- A. Design a notch filter that will only allow a range of frequencies to pass below 1kHz (1,000Hz) and above 30kHz (30,000Hz). Assuming that both the resistors have values of 10k Ω , calculate the values of the two capacitors required. Provide the calculations in box below: (2)

$$f_L = \frac{1}{2\pi R_2 C_2}$$

$$C_2 = \frac{1}{2\pi R_2 f_L} = \frac{1}{2\pi (10k)(1k)}$$

$$C_2 = 15.92 \text{ nF}$$

$$f_H = \frac{1}{2\pi R_1 C_1}$$

$$C_1 = \frac{1}{2\pi R_1 f_H} = \frac{1}{2\pi (10k)(30k)}$$

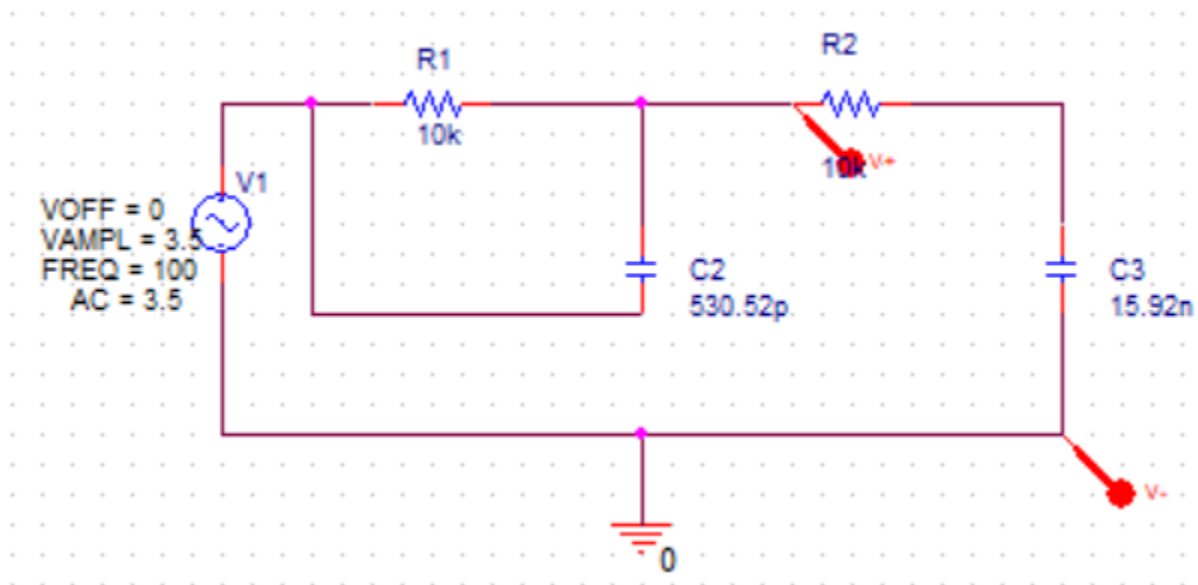
$$C_1 = 530.52 \text{ pF}$$

- B. Calculate the center frequency notch filter. (0.5)

$$f_c = \sqrt{f_L * f_H} = \sqrt{(1k) \cdot (30k)}$$

$$f_c = 5.48 \text{ kHz}$$

- C. Simulate the circuit in Figure 5 on PSpice using above calculated values. (2.5)



- D. Test the circuit by giving Input signal of $V_{in}=3.5V$ for the frequencies mentioned in table 3. Observe and record the output voltages across R_2 on below mentioned frequencies. You can use any method

of simulation to perform this task, doing AC sweep saves time. Do not forget to attached snippet of your circuit and the plot. (5)

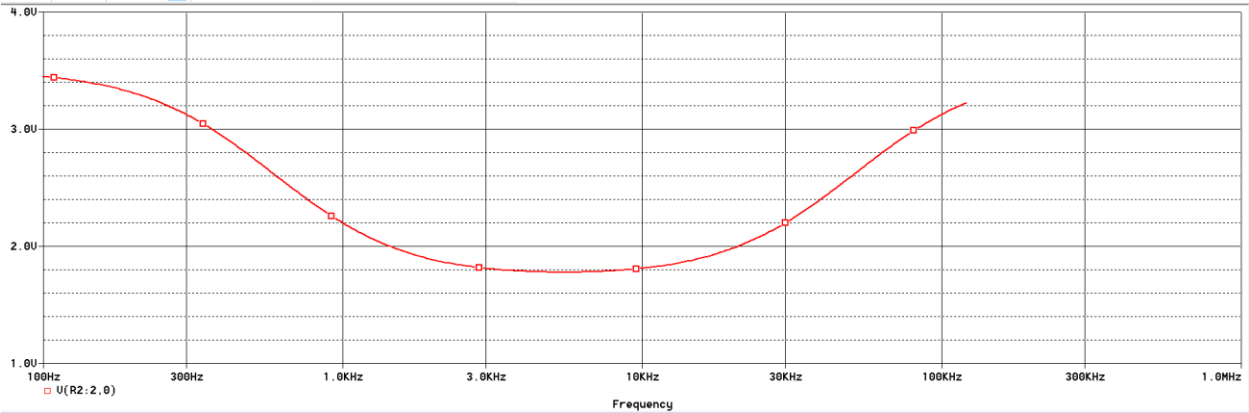
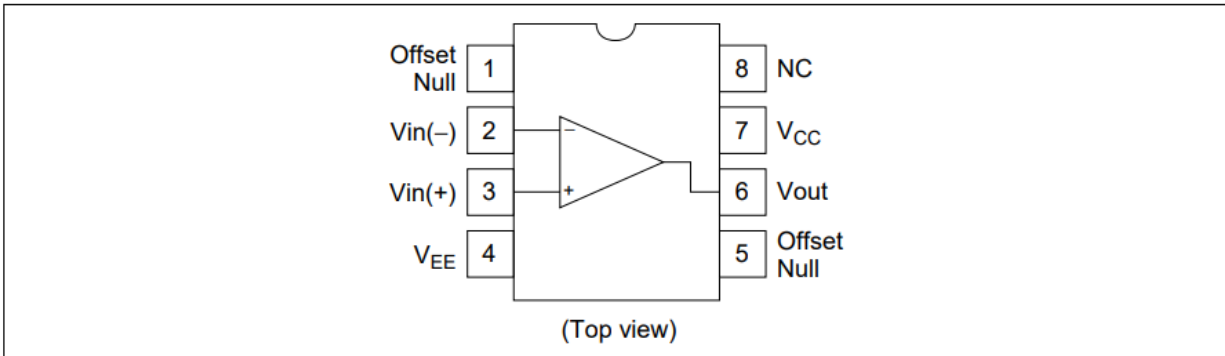


Table 3: Data Table for Notch Filter

Frequency (Hz)	V_{out}
100	3.4480 V
1k	2.1995 V
5k	1.7793 V
15k	1.8866 V
30k	2.2006 V
75k	2.9379 V
100k	3.1265 V

- Datasheets:
 - HA17741:

Pin Arrangement

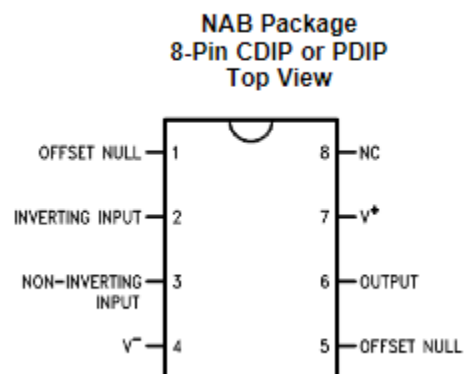


Absolute Maximum Ratings (Ta = 25°C)

Item	Symbol	Ratings		Unit
		HA17741PS	HA17741	
Power-supply voltage	V _{CC}	+18	+18	V
	V _{EE}	-18	-18	V
Input voltage	Vin	±15	±15	V
Differential input voltage	Vin(diff)	±30	±30	V
Allowable power dissipation	P _T	670 *	670 *	mW
Operating temperature	Topr	-20 to +75	-20 to +75	°C
Storage temperature	Tstg	-55 to +125	-55 to +125	°C

Note: These are the allowable values up to Ta = 45°C. Derate by 8.3 mW/°C above that temperature.

- LM741:



6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾⁽²⁾⁽³⁾

		MIN	MAX	UNIT
Supply voltage	LM741, LM741A		±22	V
	LM741C		±18	
Power dissipation ⁽⁴⁾			500	mW
Differential input voltage			±30	V
Input voltage ⁽⁵⁾			±15	V

Blank Page (Extra Space for working)